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The effect of filler type (tungsten carbide, zinc oxide) and content on the mechanical and wear behavior of jute/flax reinforced epoxy hybrid composites: Experimental and artificial neural network analysis

Mehmet Emin Demir 

Batman University, Beşiri Organized
Industrial Zone Vocational School,
Batman, Turkey

Correspondence

Mehmet Emin Demir, Beşiri Organized
Industrial Zone Vocational School,
Batman University, Batman, Turkey.
Email: memin.demir@batman.edu.tr

Abstract

The application of natural fiber-reinforced composite (NFRC) is growing rapidly in various industrial sectors, such as automotive, packaging, and construction materials, because of its affordability and excellent mechanical properties. This study investigates the effects of tungsten carbide (WC) and zinc oxide (ZnO) filler content on the mechanical and wear performance of jute/flax-reinforced hybrid composites. WC filler enhanced the tensile stress of composites more effectively than ZnO filler. Adding 1% WC and 1.5% ZnO enhanced the tensile strength of the unfilled composite by 41% and 24%, respectively. Wear tests conducted under various parameters revealed that increasing the load reduced the coefficient of friction (COF) but increased the wear rate and mass loss for both neat and filled composites. The lowest COF values were observed in WC-filled composites. It was observed that ZnO and WC-filled composites exhibited less matrix deformation, pitting, and porosity compared to neat composites. Impact resistance improved with the addition of ZnO and WC fillers, with the impact strength of 1.5% WC-filled hybrid composites increasing from 4.5 to 10.3 J/m². The experimental wear rate and COF closely matched the predicted values acquired from the model, demonstrating the success of the artificial neural network (ANN) model in precisely estimating COF and wear rate values.

Highlights

- The addition of fillers improved the mechanical properties of hybrid composites.
- Exceeding optimum filler ratio negatively affects the mechanical properties.
- Adding fillers has improved the wear resistance of jute/flax composites.
- Artificial neural network model strongly correlates with experimental results.
- Infrared spectroscopy analysis validated the interaction between filler and epoxy.

KEYWORDS

ANN, jute/flax, mechanical properties, WC, wear, ZnO

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1 | INTRODUCTION

With the advancement of materials science, there is a growing interest in natural fibers (NF) like sisal, jute, and flax, which are biodegradable, recyclable, and environmentally friendly, to meet societal needs.^{1,2} Thanks to their abundance, low density and cost, and high wear and corrosion resistance, NFs have gained the interest of engineers and scientists as an alternative to synthetic fibers in polymer matrix composites.^{3–6} Due to issues such as worker health concerns, ecological problems, energy requirements, and pollution associated with the production of synthetic fibers, reducing reliance on petroleum-based fibers makes the use of NFs more attractive.^{7–9} The biodegradability of fibers utilized in NFRC, lower environmental impact, and non-carcinogenic nature concerning human health further enhance the significance of NF.^{10,11} Thanks to the aforementioned positive attributes of NFs, they are utilized in different industrial sectors, including railways, automotive, aerospace, construction, and packaging sectors.^{12–14} In recent years, automotive manufacturers in Europe have increasingly utilized NFRC in vehicle parts such as seat backs, headliners, package trays, door panels, and dashboards.¹⁵ In fiber-reinforced composites (FRC), fiber distribution, fiber-matrix interfacial bonding, and uniform distribution of fillers have considerable effects on the properties of NFRC.¹⁶ Nanoparticles are incorporated to improve electrical, mechanical, tribological, and thermal properties of NFRC.^{17,18} In filled composites, nanoparticle content is typically kept between 1% and 10%.¹⁹ Micro or nanoparticles such as boron carbide (B_4C), silica (SiO_2), ZnO, titanium dioxide (TiO_2), and WC are employed as fillers in filled composites to advance their mechanical and tribological properties.^{20–22} Among inorganic fillers, ZnO filler is extensively utilized in numerous sectors because of its price and superior physical properties, as well as its potential to strengthen thermal, mechanical, and chemical properties of FRC.^{23,24} WC filler is a preferred material in polymer matrix composites owing to its hardness, high melting point, and outstanding physical, tribological, and mechanical properties.^{25,26} The extensive surface area of nanoparticles significantly influences the properties of composites.¹⁶ In addition, improving the weak interface adhesion between fiber/matrix, thereby preventing crack formation and stress accumulation, can be achieved through the use of filler particles.²⁷ After a certain period of use, materials in contact may experience detachment, causing the materials to lose their functionality. Another function of the hard fillers added to FRC is to distribute evenly between the matrix and fibers, forming a strong bond and reducing the contact area, thereby enhancing the tribological properties.²⁸

Researchers carried out numerous researches on NFRC and filled composites.^{29–34} Chand and Dwivedi investigated the hardness and wear characteristics of chemically treated and untreated jute-reinforced composites. It was found that the tensile stress and hardness of treated composites were improved, while the wear volume was greater in untreated jute fiber-reinforced composites (JFRC). SEM analysis has revealed that due to poor matrix-fiber bonding, unmodified fibers easily separate from each other, which in turn reduces the strength.³⁵ Li et al. noted that the tensile stress of flax FRC increased with up to 20% flax content, but beyond this ratio, a decreasing trend was observed. They also discovered that tensile elongation decreased as the fiber content increased.³⁶ Athith et al. examined the wear and mechanical behavior of FRC by incorporating different amounts of WC particles into jute-sisal-glass FRC. They discovered that an increase in WC filler ratio led to improvements in impact, flexural, and tensile strengths. They also reported that 5% WC filler enhanced the tensile stress of the jute composite from 61.82 to 82.68 MPa.³⁷ Reddy et al. examined the impact of nano WC fillers on the erosive wear performance and hardness of matrix composites. They observed that the composite with 1% WC filler exhibited the highest hardness among the epoxies with WC content ranging from 1% to 4%. They also noted that WC filler improved erosion wear resistance, with the best results observed at a 2% WC filler content. With the addition of WC, due to its hardness, the ductile structure of the produced composite transitioned into a semi-ductile form.³⁸ Mohan et al. examined the erosive wear performance of WC-filled composites. They concluded that the erosion rates of the neat glass fiber reinforced composites (GFRC) were higher. SEM analyses revealed that matrix deformation and fiber detachment were more pronounced in the neat composites. The lower erosive wear of the WC-filled composite has been associated with the lower fiber-matrix debonding in the filled composites.³⁹ Kadhim et al. aimed to enhance the wear resistance of thermoset polymers using different fillers. They compared the wear resistance of polymers by incorporating varying amounts of WC and SN. They found that the WC-filled composites increased the hardness values more significantly. Additionally, they noted that the wear resistance of the epoxy composites improved further with the WC filler, and COF values were obtained to be lower. They explained that the primary reason for the increased hardness resistance of the polymer is the ability of WC and SN particles to hinder crack propagation within the epoxy matrix.⁴⁰ Braga and Magalhaes focused on investigating the effects of the jute and glass fiber ratios in epoxy resin-based jute/glass FRC on the composites density, impact, tensile, flexural, and

thermal properties. It was observed that an increase in the glass fiber content within the jute/glass hybrid composite resulted in enhanced tensile strength and density, thanks to the powerful interfacial adhesion between the fiber and epoxy. They also noted that reducing jute fiber content from 25% to 18% increased the impact strength from 3.53 to 5.49 J.⁴¹ Bahadur and Sunkara investigated the tribological performance of four different nanofillers (ZnO, TiO₂, SiC, and CuO) by incorporating them into a polyphenylene sulfide (PPS) matrix to enhance its wear resistance, given its inherently low wear performance. The COF of the unfilled PPS matrix was 0.43, while the addition of 1% ZnO filler reduced the COF to 0.39.⁴² Sathishkumar et al. utilized ZnO filler to enhance the impact and compressive stress of JFRC. They reported that the addition of ZnO particles enhanced the compressive stress to 27.71 MPa and the impact strength to 8.12 kJ/m².⁴³ A review of the literature shows that both experimental methods and the use of experimental data with ANN for data processing are employed to predict the wear performance of composites. This approach helps evaluate the wear performance of materials while reducing the number of experiments, saving time, and cutting costs.^{44–49}

With the growing environmental awareness and consciousness, NFs are being increasingly studied in scientific research. The production of new hybrid materials, coupled with the appropriate selection of fibers and fillers, enhances the potential use of these materials in industries such as construction, packaging, and vehicle industries. Although investigations have been performed on the use of fillers in NFRC, the limited research on the tribological and mechanical behavior of WC and ZnO particles in NFRC has been the primary motivation for

this study. This study used eco-friendly jute and flax fibers to produce polymer matrix-based hybrid composites filled with nano WC and ZnO. The effects of filler particles and filler content on the mechanical properties and wear performance of the composites were investigated. After tensile testing, the fracture surfaces and wear testing, the worn surfaces were analyzed using SEM images.

2 | MATERIALS AND METHODS

2.1 | Materials

ARC 152 epoxy resin and W152 hardener were sourced from ARC Marin Corporation, Kocaeli, Turkey. The hardener and epoxy resin mixture ratio was made according to catalog specifications, with a 1:4 ratio. Jute fibers, used as reinforcement material, were procured from Uysallar leather, İstanbul, Turkey, and flax fabric was sourced from Macit upholstery fabric, İstanbul, Turkey. The plain woven jute and flax fabrics weigh 250 and 200 g/m², respectively. WC and ZnO filler particles were purchased from Nanografi company, Ankara, Turkey. The filler and fiber properties and composite laminate composition are provided in Table 1.

2.2 | Production of filled-hybrid composites

Both neat and filled hybrid composites were produced using the hand lay-up method with a 6-layer JFFFFJ (J:

TABLE 1 The physical properties of the filler particles, fibers¹⁴ and composite laminate composition.

Filler	Particle size (nm)	Density (gr/cm ³)	Crystal structure	Specific surface area (m ² /g)	
WC	55	15.7	spherical	1.5–2	
ZnO	18	5.5	spherical	25–65	
Fiber	Elongation (%)	Elastic modulus (GPa)	Density (g/cm ³)	Tensile strength (MPa)	
Flax	2.7–3.2	27.6	1.5	500–1500	
Jute	1.5–1.8	26.5	1.3	393–773	
wt.%	Jute + Flax	Epoxy	WC	ZnO	
Neat	50	50	-	-	
0.5 WC	50	49.5	0.5	-	
1 WC	50	49	1	-	
1.5 WC	50	48.5	1.5	-	
0.5 ZnO	50	49.5	-	0.5	
1 ZnO	50	49	-	1	
1.5 ZnO	50	48.5	-	1.5	

Jute, F: Flax) structure. The required filler particles were accurately weighed using a precision scale and then mixed into the epoxy matrix, which was stirred in an ISOLAB-branded ultrasonic bath for 30 min to provide uniform distribution of the fillers within the epoxy matrix. The epoxy/hardener mixture was applied evenly to the surfaces of 30 cm × 30 cm jute and flax fabrics using a roller, producing neat plates with a JFFFFJ layer structure. The same process was then repeated with epoxy composites containing WC and ZnO fillers, resulting in the production of a total of 7 plates. All produced plates were left to cure at room temperature for 24 h without applying any pressure. After curing, tensile, impact, and wear samples were extracted from the plates following ASTM standards.

2.3 | Tensile test

Tensile tests of the 250 × 25 × 4 mm samples, cut according to ASTM D3039 standards, were performed on a SHIMADZU AG-IC device with a capacity of 250 kN at 25°C and a 1 mm/min test speed. Three tensile tests were conducted for seven hybrid composites, and the average of the results was taken.

2.4 | Flexural test

Flexural tests of the 125 × 12.7 mm samples, cut according to ASTM D790 standards, were performed on a SHIMADZU AG-IC device with a 1 mm/min test speed. Flexural stress was calculated with the formula given in Equation 1.

$$\sigma_f = 3PL/2bd^2 \quad (1)$$

2.5 | Wear and impact test

Wear tests were conducted under dry sliding conditions on samples cut to 25 × 40 mm dimensions utilizing a Turkeyus branded wear device, following ASTM G-99 standards. The wear volume was computed based on the trace width, and formulas used are provided in Equations (2), while the wear rate formulas are provided in Equation (3).⁵⁰ Before and after the wear tests, the composites were weighed using a precision scale with a sensitivity of 0.0001 g to calculate the wear loss. A steel ball with a 6 mm diameter was used in the wear tests. The effects of load (3, 5, 8 N), sliding distance (100, 200,

300 m), and filler content (0.5, 1, 1.5 wt. %) on the wear rate and COF were examined. A computer data logger acquired COF values. Wear disk speed was set at 250 rpm, and the track width was adjusted to 10 mm. In the Charpy impact test, the samples were prepared by ASTM D6110 standards, and experiments were performed with an MT3016 Probador testing device.

$$\text{Wear volume} = 2 * \pi * R [r^2 * \sin^{-1}(a/2 * r) - (a/4) * (4 * r^2 - a^2)]^{1/2} \quad (2)$$

Here, R implies wear track radius, r denotes ball radius, and a represents trace width.

$$\text{Wear Rate} = \text{Wear volume (mm}^3\text{) / Sliding distance (m)} \quad (3)$$

Here P represents the applied load, σ_f denotes the flexural stress, L is the distance between the supports, b is the sample width, and d refers to the sample thickness.

2.6 | Artificial neural networks

ANNs are systems inspired by the learning and reasoning techniques of neurons in the biological nervous system. They aim to solve various problems without the need for excessive experimentation.⁵¹ Wear is a complex phenomenon that is difficult to define through mathematical modeling. For this purpose, ANNs have been employed to study the wear behavior of composites, aiming to uncover relationships between process parameters and identify the best wear performance characteristics.⁵² In the ANN model, a multilayer, backpropagation, and feed-forward network was utilized. The Levenberg–Marquardt (LM) algorithm was applied due to its efficiency as a backpropagation method for solving nonlinear least-squares problems. It is optimized to minimize the sum of squared errors.⁵³ The model consists of four inputs (filler ratio, type, load, and slide distance), three layers, and one output. To assess the validity of the developed ANN model and prevent overfitting, a cross-validation technique known as K-fold was implemented. The dataset was randomly divided into five parts, and this process was repeated five times, using each part as the test data once. The ANN network was created using MATLAB 2023b software.

3 | RESULTS AND DISCUSSION

3.1 | Attenuated total reflectance-fourier transform infrared spectroscopy (ATR-FTIR) analysis

The chemical structure of Neat, WC-filled, and ZnO-filled are studied by ATR-FTIR. The spectra presented in Figure 1 demonstrate that neat displays a characteristic peak of hydrocarbon (C—H) stretching at $2850\text{--}2970\text{ cm}^{-1}$, C—O absorption at $1050\text{--}1300\text{ cm}^{-1}$, and epoxy group at $830\text{--}912\text{ cm}^{-1}$.⁵⁴ When the infrared spectroscopy of WC-filled composites was analyzed, sharp peaks were observed at 860, 2840, and 2940 cm^{-1} , revealing the presence of reaction bands. The surface coating of the WC-filled composite is validated by bands observed here. Bands at 2840 and 2940 cm^{-1} indicate the presence of the alkyl group, which is associated with the WC-filled composites. The band at 860 cm^{-1} is a result of a reaction between the methoxy group of epoxy and WC particles.⁵⁵ In addition, the ZnO/Neat composite exhibited an additional Zn—O characteristic peak at 516 cm^{-1} compared to the neat sample, pointing to a strong chemical interaction, including potential chemical bond formation with the epoxy amine chain with ZnO particles.^{56,57} This confirms that the epoxy is effectively surface-treated with WC and ZnO fillers.

3.2 | Tensile test results

The experimental failure load–displacement curves for both neat and filled composites and standard deviations of tensile results are presented in Figure 2. Figure 2

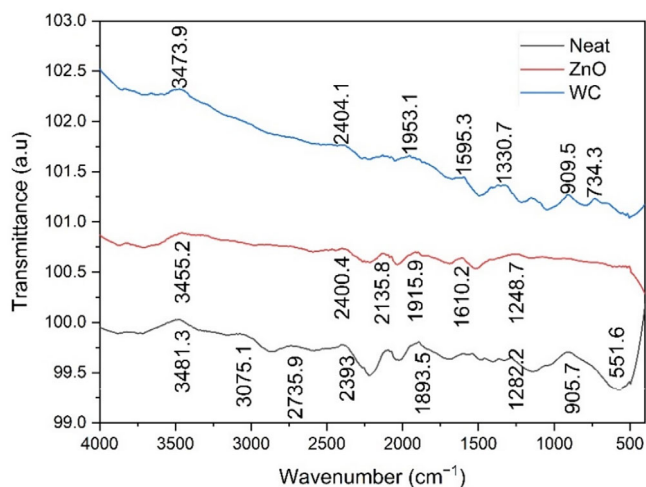
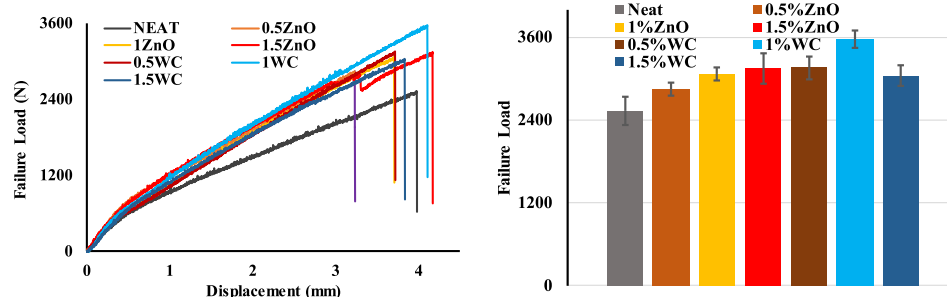


FIGURE 1 ATR-FTIR profile of (A) Neat, (B) WC-filled, and (C) ZnO-filled.

demonstrates that both neat and filled composites exhibited brittle fracture behavior upon failure. Among the tested specimens, the highest failure load values were recorded for the 1% WC-filled flax-jute composites, while the lowest values were recorded for neat flax-jute composites. A 12.34% increase in failure load values was observed for the 0.5% ZnO-filled flax-jute hybrid composites compared to the neat flax-jute hybrid composites. Similarly, 1% ZnO filler resulted in a 21.29% increase, while 1.5% ZnO filler led to a 24.38% increase in failure load values relative to the neat flax-jute hybrid composites. These results indicate that the increased ZnO filler ratio in flax-jute composites enhances their failure load capacity. The increase in tensile strength in ZnO-filled composites can be attributed to the enhanced load-bearing capacity resulting from the increased adhesion strength at the matrix-fiber interface. Zaman et al. stated that the addition of nano ZnO particles to the epoxy matrix enhances tensile strength by forming strong hydrogen bonds with the OH groups in the resin, thereby improving the interaction between the epoxy and the filler.⁵⁸ The low standard deviation of the tensile test results of the composites indicates that the composites exhibit homogeneous behavior with respect to mechanical properties and that the manufacturing process is consistent and repeatable. In a related study, Wu et al. fabricated biocomposites filled with nano-ZnO and reported that incorporating 12.4% nano-ZnO improved the tensile strength of the biocomposites by 38%.⁵⁹ Devarajua et al. reported an increase in tensile strength, reaching 39.44 MPa, with the incorporation of 0.5% nano-ZnO into NaOH-pretreated polymer composite materials.⁶⁰ Hawkins et al. reported a 20% enhancement in the tensile stress of epoxy composites filled with ZnO-modified carbon nanotubes. This improvement in tensile strength was ascribed to the homogeneous dispersion of ZnO-modified particles within epoxy, facilitating strong interfacial bonding with the epoxy and enabling effective stress transfer.⁶¹ El-Ebissy et al. stated that the mechanical properties improved by adding up to 1% ZnO filler. However, a further increase in filler ratio to 2% resulted in degraded mechanical properties, as particle agglomeration within the epoxy caused an increase in brittleness.⁶² Failure load values of flax-jute hybrid composites increased by 24.7%, 41.04%, and 20% with the incorporation of 0.5%, 1%, and 1.5% WC filler. Increasing the WC filler ratio from 0.5% to 1% enhanced the tensile stress by 13%, raising it from 3156 to 3570 N. The primary reason for the enhancement in the tensile stress of the composite with WC filler is the uniform dispersion of WC within epoxy, which enhanced load transfer capability and consequently improves tensile strength. However, at a 1.5% WC filler content, a reduction in tensile stress was

FIGURE 2 Influence of filler type and ratio on failure load–displacement and standard deviation of tensile test results.



observed. The reduction in tensile stress is due to the clustering of particles in the matrix, resulting in stress concentration regions. In another study where WC filler was used in FRC, it was found that WC filler enhances tensile stress at low filler contents. However, at higher filler contents, inadequate wetting of WC by the matrix leads to a poor interface bond between the fiber/matrix, which negatively affects tensile strength.⁶³ Reddy et al. noted that the best mechanical performance was achieved in composites with 2% WC filler content, within a range of 1%–4% filler. Beyond this proportion, a decline in mechanical properties was observed.²⁶ Sanjay et al. noted that the incorporation of WC particles and its increase in jute-sisal-E-glass composites enhanced both flexural and tensile stress. They indicated that the filler particles strengthened the fiber-matrix interfacial bond.³⁷ The highest displacements were recorded in 1% WC-filled flax-jute hybrid composites, while the lowest displacements were observed in 0.5% ZnO-filled flax-jute hybrid composites. Due to the brittle nature of failure in WC and ZnO-filled flax-jute hybrid composites, reductions in displacement were noted compared to neat flax-jute hybrid composites. Nevertheless, no significant increases in displacement were observed in 1.5% ZnO and 1% WC-filled flax-jute hybrid composites. Wu et al. reported that in ZnO-filled composites, elongation initially decreased with increasing filler content, followed by a trend of increase.⁵⁹

3.3 | Flexural test results

Figure 3 illustrates three-point flexural results of hybrid composites, varying according to filler content and type. It was observed that adding fillers enhanced flexural stress of composites. The highest flexural strength was achieved for the 1% ZnO-filled composite at 70.5 MPa and the 1% WC-filled composite at 72.81 MPa. In composites under flexural stress, crack formation initiates in epoxy and is transferred to nanoparticles. The hard WC and ZnO particles reduce crack formation, thereby enhancing flexural performance. Moreover, the strong

bond formation between the matrix/filler/fiber enhances load transfer capability, thereby improving flexural stress of composites. Increasing the filler content to 1.5% led to a slight decline in the flexural stress of the ZnO and WC-filled hybrid composites. With an increased filler ratio, the agglomeration effect of the nanoparticles disrupts matrix continuity, weakening the bond between the fiber/matrix/filler. Another reason for the decrease in flexural strength at high filler concentrations is the reduction in chain length within the epoxy matrix as the filler content increases.²⁶ In a similar study, it was reported that ZnO filler increased the flexural resistance in GFRC composites, and this trend continued up to a 3% ZnO filler content. However, exceeding 3% ZnO content resulted in a reduction in flexural stress owing to agglomeration, which prevented adequate wetting of fillers by the epoxy.⁶⁴ Joshi et al. reported that in basalt-filled composite, the best flexural performance was achieved with a ZnO filler content ranging from 1% to 5%, specifically at a 2% filler ratio. However, with increased filler content, a reduction in flexural stress was observed, attributed to the formation of a heterogeneous structure, resulting from the reduced wetting ability of the matrix.⁶⁵ In micro WC-filled HDPE composites, it has been highlighted that there was an improvement in flexural stress up to a filler content of 12.5%. However, beyond this ratio, the strength was adversely affected owing to poor adhesion between filler/matrix.⁶⁶

3.4 | Fracture morphology of tensile samples

Figure 4 presents SEM images of the fractured surfaces of WC-filled jute-flax hybrid composites. As shown in the tensile graph, the separation of fibers from the matrix occurs frequently in brittle matrix composites.⁶⁷ Fiber separations observed in SEM analyses and sudden load drop ruptures seen in tensile graphs support the brittle composite structure. Under the applied tensile loads, various failure mechanisms were observed, including kinks, fiber breakage, dispersion of fibers in multiple directions,

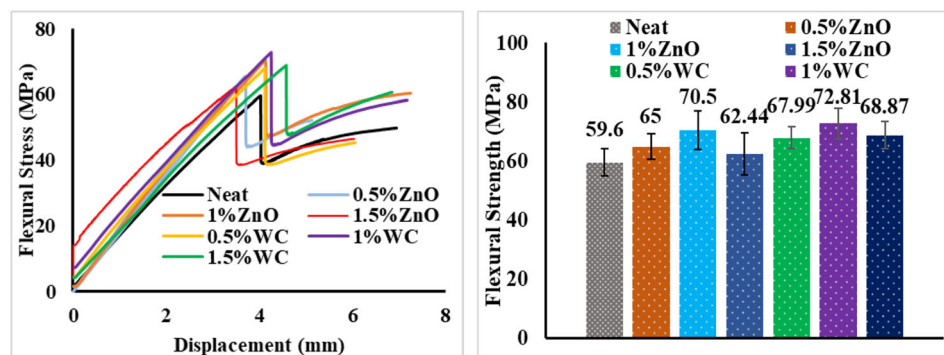


FIGURE 3 Flexural strength of neat and filled composites.

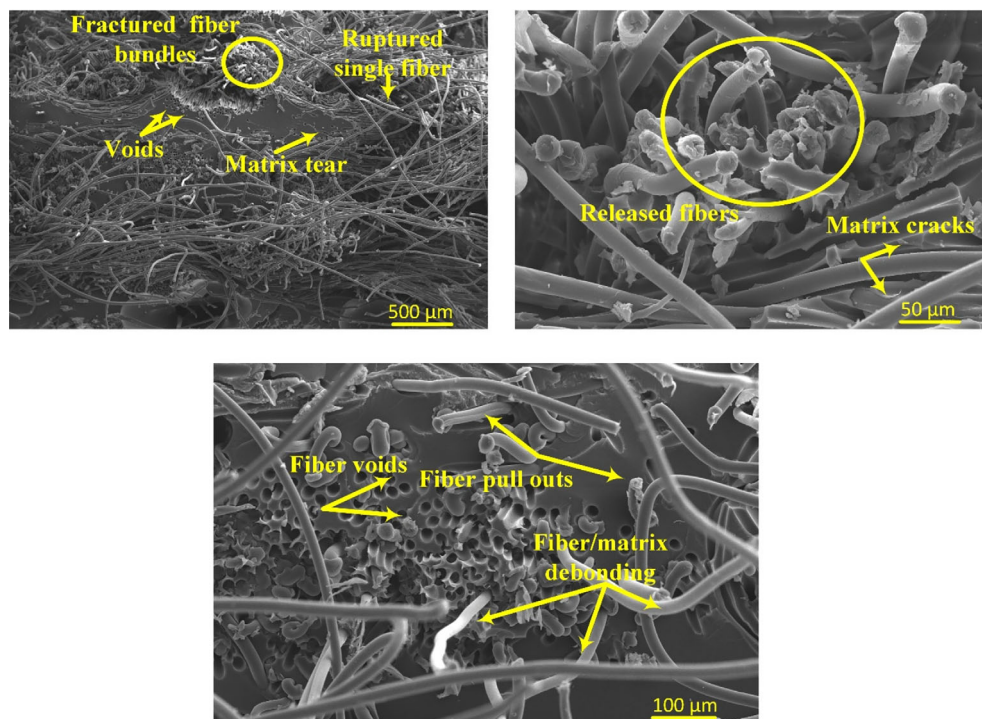


FIGURE 4 Fracture morphology of 1.5% WC-filled hybrid composite.

bundled and discrete fractures, and fiber pull-outs from the matrix material. In certain matrix regions, fibers are fractured in dense bundles, indicating strong interfacial adhesion in those areas. Matrix damages and fiber separation from the matrix observed in SEM analyses were effective in the initiation, propagation, and continuation of cracks until the composite fractured. Moreover, in the case of high deformation energies, the composite surfaces appear with cracks in SEM analyses.⁶⁸ The attainment of high yield strength values in tensile tests further corroborates the formation of robust adhesion between fiber and epoxy. Moreover, uniform dispersion of fillers within hybrid composites emerges as a critical factor in enhancing the tensile strength. Conversely, in other regions, scattered fiber fractures accompanied by significant voids highlight areas of weak interfacial bonding. These observations underscore the heterogeneity in the bonding quality across the composite material.

The production conditions, particle agglomerations within the epoxy matrix, and nonlinear stress distributions were identified as the primary causes of the weak adhesive bonds observed. Especially in hybrid composites with high nanoparticle content, the reduction in tensile stress was ascribed to the presence of fibers that were not fully covered by the epoxy matrix.⁶⁹ The smooth surfaces observed in SEM analyses indicate insufficient epoxy material in those areas or that the epoxy is curing too quickly.⁷⁰ The presence of intense fiber fractures in regions near the matrix and dispersed fiber fractures further away from the matrix suggests that both the matrix material and fibers were resistant to tensile loading and that their compatibility was adequate. As a result of the tensile testing, the fractured fibers increased with the initiation of cracks. Since the elongated and curled fibers tend to flatten due to the high stresses formed in the composite under tensile loads, micro cracks initiate and

spread in the composite.⁷¹ In addition to crack propagation, the decrease in potential energy also facilitated the formation of new fiber surfaces. The elongation and fracture behavior of fibers in the composite indicated that the orientations of these fibers remained unaltered during the composite manufacturing process, and they exhibited a tendency to fracture along the direction of applied tensile forces. Additionally, matrix deformations, including fiber pull-outs, voids, and cracks, were evident in the SEM analysis.

3.5 | Wear test results

COF, wear rate, and mass loss of neat composites with varying ZnO and WC filler contents were compared under different wear parameters. Figure 5 presents the COF values and corresponding COF graphs obtained from the wear device, with variations in COF based on filler content, load, and sliding distance.

Figure 5A illustrates the effect of load variation on COF. With a constant sliding distance, it is evident that as the load increased from 3 to 8 N, COF values decreased for both neat and filled composites filled with 1.5% ZnO and 1.5% WC. The lowest COF values were recorded under an 8 N load, with 0.22 for WC-filled composites, 0.28 for ZnO-filled composites, and 0.37 for neat composites. It is evident that the addition of filler to the hybrid composites successfully reduced the COF values. Across all loads, ZnO-filled hybrid composites exhibited

higher COF values compared to WC-filled composites. The friction between the contacting parts causes a rise in temperature at contact points, causing the polymer matrix to soften and alter its viscoelastic properties. The increase in load, along with the rising temperature at the contact area, promotes the formation of a thin lubricating layer, resulting in a decrease in COF values.^{72,73} The equation that expresses the COF value of polymeric materials concerning load is $\mu = kN^{(n-1)}$.⁷⁴ In this context, N represents the applied load, and μ represents COF. As indicated by the formula, there is an inverse relationship between the applied load and the COF value. Sudhagar and Kumar attributed the decreasing trend in the COF with increasing load in FRC to the reduction in shear resistance at higher loads.⁷⁵ In line with our results, Şahin and Baets indicated that the COF decreased with increasing load in FRC, attributing this behavior to thermal softening and the reduction of the plowing component.⁷⁶ Kumar and Anand reported that in ramie FRC, COF initially increased with increasing load and then followed a decreasing trend. The primary factor behind this behavior was the rising temperature in the wear region, which softened the epoxy matrix, formed a tribo film on the opposing surface, and reduced the contact between the counterface and fibers.⁷⁷ However, once the critical load threshold is exceeded, the polymer chain begins to weaken, causing an increase in COF value. This behavior reflects the breakdown of the polymer structure under excessive load, which causes a rise in friction.⁷⁸ Yallow et al. stated that in jute-reinforced epoxy composites, as

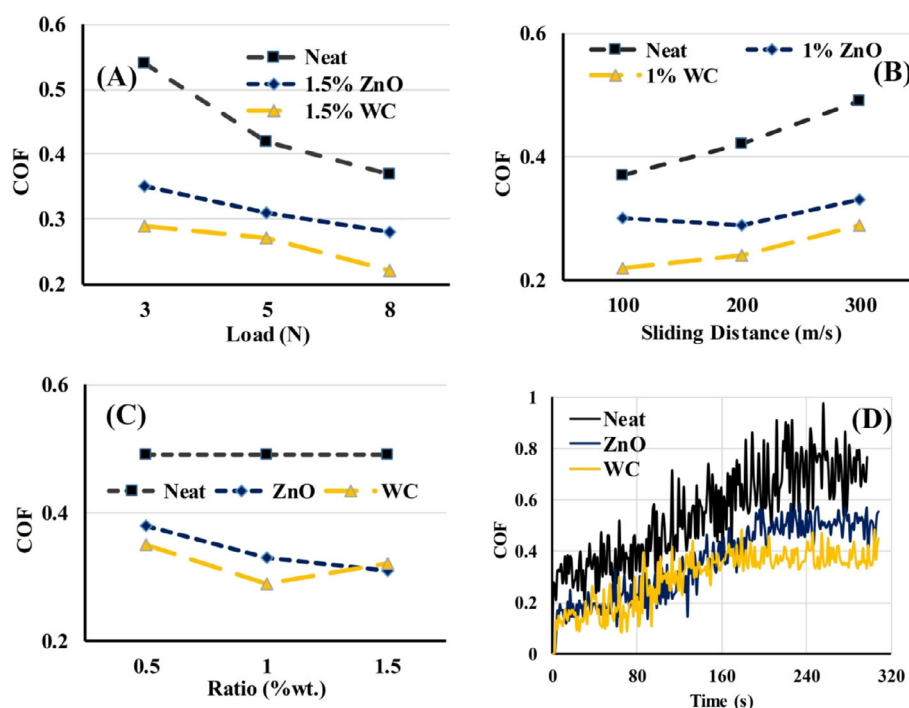
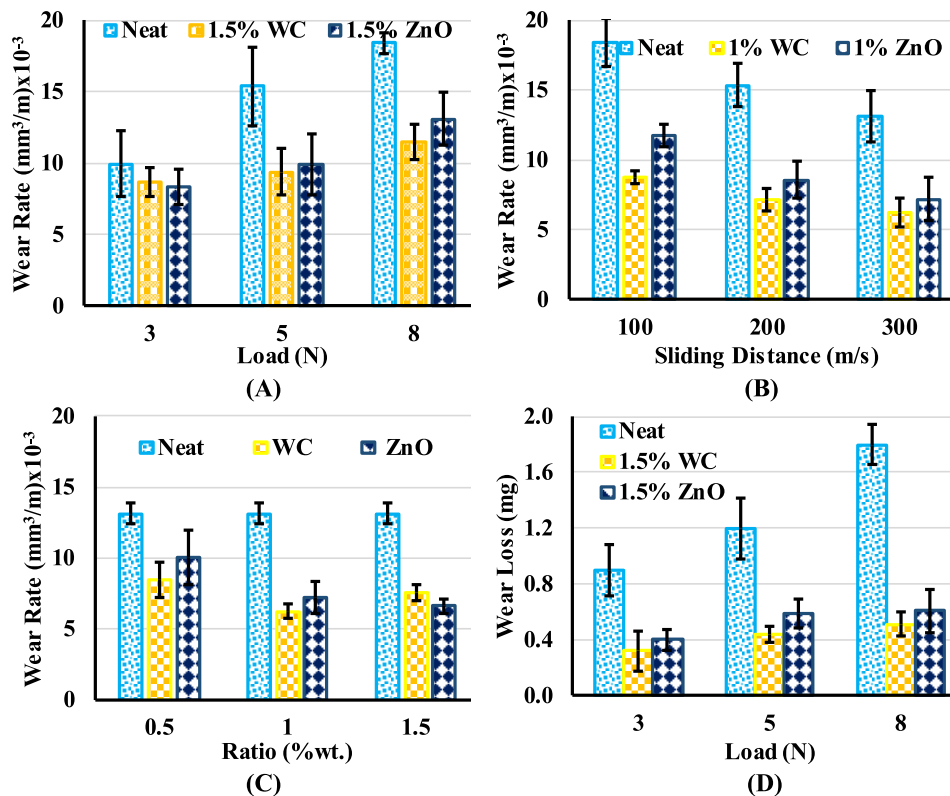


FIGURE 5 COF depending on (A) load at 200 m sliding distance, (B) sliding distance at 5 N load, (C) ratio, 5 N load 300 m sliding distance, (D) time.

the applied load increases, a thin polymer film forms over the fiber layer, which helps to reduce the COF value. This phenomenon is due to the formation of a lubricating layer that decreases friction between surfaces, resulting in a lower COF as the load increases.⁷⁹ The decrease in COF with the increasing load has also been observed in similar studies.^{80,81} Figure 5B illustrates the impact of sliding distance on COF. Tests conducted at sliding distances between 100 and 300 m with a 5 N load revealed a nonlinear increase in COF values for both neat and filled composites as the sliding distance increased. With a sliding distance of 300 m, the COF value of the neat composite was 0.49, while the COF value of the 1% WC-filled composite was 0.27. In the 1% ZnO-filled composites, the COF value increased from 0.30 to 0.33 as the sliding distance increased from 100 to 300 m. The observed increase in COF values with the rising sliding distance in all three composite types can be associated with the rise in the composite surface temperature, which leads to the thermal deterioration of the epoxy. Song et al. reported that nano-ZnO particles enhanced the wear performance of the epoxy matrix and reduced COF values. They attributed the reduction in COF to the ability of ZnO nanoparticles to decrease contact between the steel pin and epoxy. They also reported that a higher ratio of ZnO particles extended the wear life of composite.⁸² Li et al. reported that in composites reinforced with different amounts of ZnO, increasing the ZnO ratio helped to reduce wear loss and COF.⁸³ Bobbili and Madhu observed that in filled composites, the increase in sliding distance caused an increase in COF.⁸⁴ In Figure 5C, the impact of the filler ratio on COF values of hybrid composites is shown. A 1% WC filler under a 5 N load lowered the COF value of the neat composite from 0.49 to 0.29. At a 1.5% filler content, the WC-filled composite's COF value was greater compared to ZnO-filled composite. The increase in filler content is believed to result in the agglomeration of filler particles within the matrix, and the abrasive effect of these agglomerated clusters between the composite and pin likely causes the increase in COF values.⁸⁵ In Figure 5D, the COF graphs obtained from the wear device for neat, WC, and ZnO-filled composites are presented. Initially, the COF value of the neat hybrid composite was 0.27, but over time, due to surface deformations and material detachment, the COF value increased to 0.75. The COF value, which initially showed an upward trend, later stabilized and leveled off. It was found that fluctuations of the filled composites followed a smoother path compared to the neat composite. During the initial stages of wear, the COF values of the WC and ZnO-filled composites were similar. However, as the wear progressed, the WC-filled composite's COF value was lower than that of the ZnO-filled composite.

Figure 6 illustrates variations in mass loss and wear rate according to filler ratio, sliding distance, and load. In Figure 6A, the influence of load increase on the wear rate of neat, WC, and ZnO-reinforced composites is shown at a slide distance of 200 m. The wear rate increased with higher loads across all composite types. With a higher load, the penetration depth of the steel pin on the composite surface increases, and due to the effect of shear stress, the material begins to flow. When the shear force reaches a certain level, material detachment from the composite occurs, which contributes to the increased wear rate. Another factor that increases the wear rate is the temperature rise at the contact region, caused by increased load, which causes the softening of the epoxy and further increases material detachment.⁸⁶ The lowest wear rate of $8.33 \text{ mm}^3/\text{m} \times 10^{-3}$ was obtained for the 1.5% ZnO-filled composite under a 3 N load. The reason for the decrease in wear rate in filled composites is a result of the filler particles enhancing the load-bearing capability of the matrix and the formation of a tribo-surface between contacting surfaces, which helps protect composite material against wear.^{87,88} Furthermore, the filler particles activate the epoxy groups, thereby strengthening the adhesion capability between the matrix and filler.⁸⁹ As the load increases, it is observed that WC-filled composites exhibit a lower wear rate. Adding 1.5% WC filler to the hybrid composite decreased the wear rate from 18.41 to $11.45 \text{ mm}^3/\text{m} \times 10^{-3}$ under an 8 N load. At a 5 N load, the addition of 1.5% ZnO filler reduced the wear rate of the neat composite by 35%. Figure 6B illustrates changing wear rates of composites based on slide distance. With a 300 m sliding distance and a 5 N load, wear rates of neat composites, 1% WC-filled, and 1% ZnO-filled composites were $13.1 \text{ mm}^3/\text{m} \times 10^{-3}$, $7.21 \text{ mm}^3/\text{m} \times 10^{-3}$, and $6.23 \text{ mm}^3/\text{m} \times 10^{-3}$, respectively. The embedded hard inorganic WC and ZnO nanoparticles within epoxy increased bonding between fiber and matrix, thereby enhancing resistance to contact with the opposing surface, resulting in lower wear loss.^{90,91} The reduction in wear rate is because, at lower wear distances, the abrasive pin only contacts the soft matrix layer. However, at higher sliding distances, the pin reaches the filler and fiber particles within the matrix, leading to less wear. This is owing to the enhanced wear performance offered by reinforcement materials, which act as barriers to further material loss.⁹² Wear rates of neat composites were greater than filled composites at all three sliding distances, with the lowest wear rate obtained in the WC-filled composite. In Figure 6C, the impact of filler ratio on wear rate with a slide distance of 300 m is shown. For WC-filled hybrid composites, the minimum wear rate was provided at a 1% filler ratio, which was $6.23 \text{ mm}^3/\text{m} \times 10^{-3}$, while for the ZnO-filled

FIGURE 6 Wear rates depending on (A) load at 200 m sliding distance, (B) sliding distance at 5 N load, (C) ratio, 5 N load 300 m sliding distance, (D) wear loss depending on load.



composite, it was $6.61 \text{ mm}^3/\text{m} \times 10^{-3}$ at a 1.5% filler content. Increasing the WC filler content to 1.5% led to a trend of increasing wear rate, which is due to the tendency of added WC fillers to agglomerate, resulting in a non-homogeneous distribution within the matrix. The higher hardness of WC filler particles compared to ZnO filler particles enhances tribological performance more effectively, providing strong resistance against abrasive effects and reducing wear loss. Consequently, WC exhibits superior wear resistance compared to ZnO filler. Similar studies have also indicated that the agglomeration effect of fillers weakens the adhesion strength between the matrix and filler, increasing the wear rate.^{93,94} Tensile test results also confirm this observation. The WC and ZnO-filled composites exhibited better wear resistance. Adding 0.5% WC filler to the neat composite reduced the wear rate from 13.14 to $8.45 \text{ mm}^3/\text{m} \times 10^{-3}$. Saadi et al. reported that adding WC fillers at varying amounts up to 3% into the epoxy matrix improved the wear resistance of the epoxy matrix. The best wear resistance was achieved with a 2% WC filler content. They attributed the decrease in wear rate to the role of the hard WC fillers acting as a barrier between the matrix and the counter surface.⁹⁵ Nachiket et al. observed that the addition of WC filler reduced the wear rate in epoxy composites compared to neat epoxy. They also noted that increasing the WC filler content from 5% to 10% harmed wear resistance. Additionally, they reported that an increase in load increased wear loss.⁹⁶

Mohan et al. stated that in WC-filled GFRC, the WC filler and increasing the sliding distance reduced the wear loss of the GFRC, resulting in enhanced tribological performance.⁹² Similar studies have also observed that the wear rate decreases with increasing sliding distance.^{86,97} In Figure 6D, the wear losses of filled and neat composites under increasing loads are presented. It can be observed that the mass loss of neat, WC, and ZnO-filled composites increases with higher loads. As surface temperature rises with increasing load, the plasticized matrix layer loses its load-bearing capacity, leading to more matrix detachment.^{86,98} The lowest wear loss was obtained at a 3 N load in the 1.5% WC-filled composite, with a value of 0.32 mg. The absence of inorganic particles in the matrix of the neat composite makes it more susceptible to wear, resulting in increased mass loss. The lower wear loss in the WC-filled composite can be explained by the formation of a barrier between the composite and the opposing surface, with nano WC particles preventing material detachment.

3.6 | Worn surface morphology

In Figure 7, SEM images of wear samples of neat, ZnO-filled, and WC-filled hybrid composites under the same conditions are presented. In Figure 7A, the worn surface of the neat composite shows significant matrix deformation and detachment. Severe cavity and pit formations

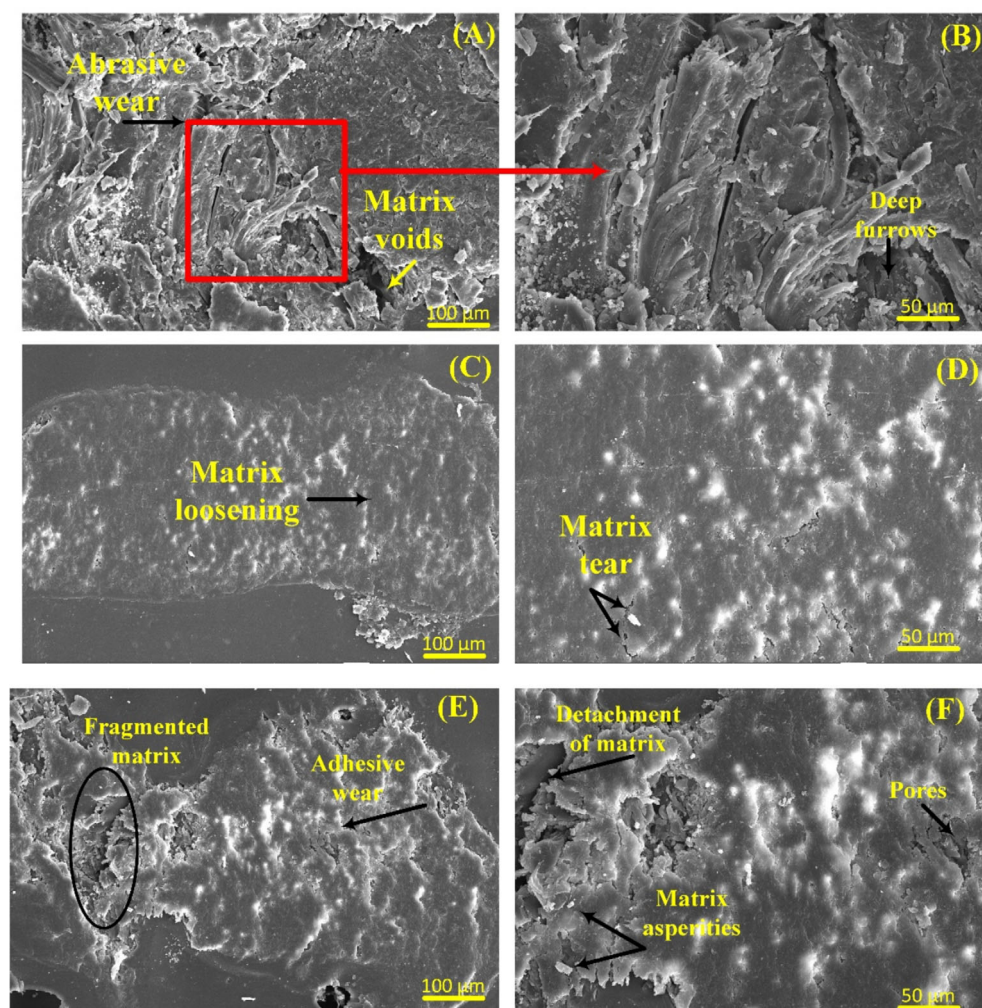


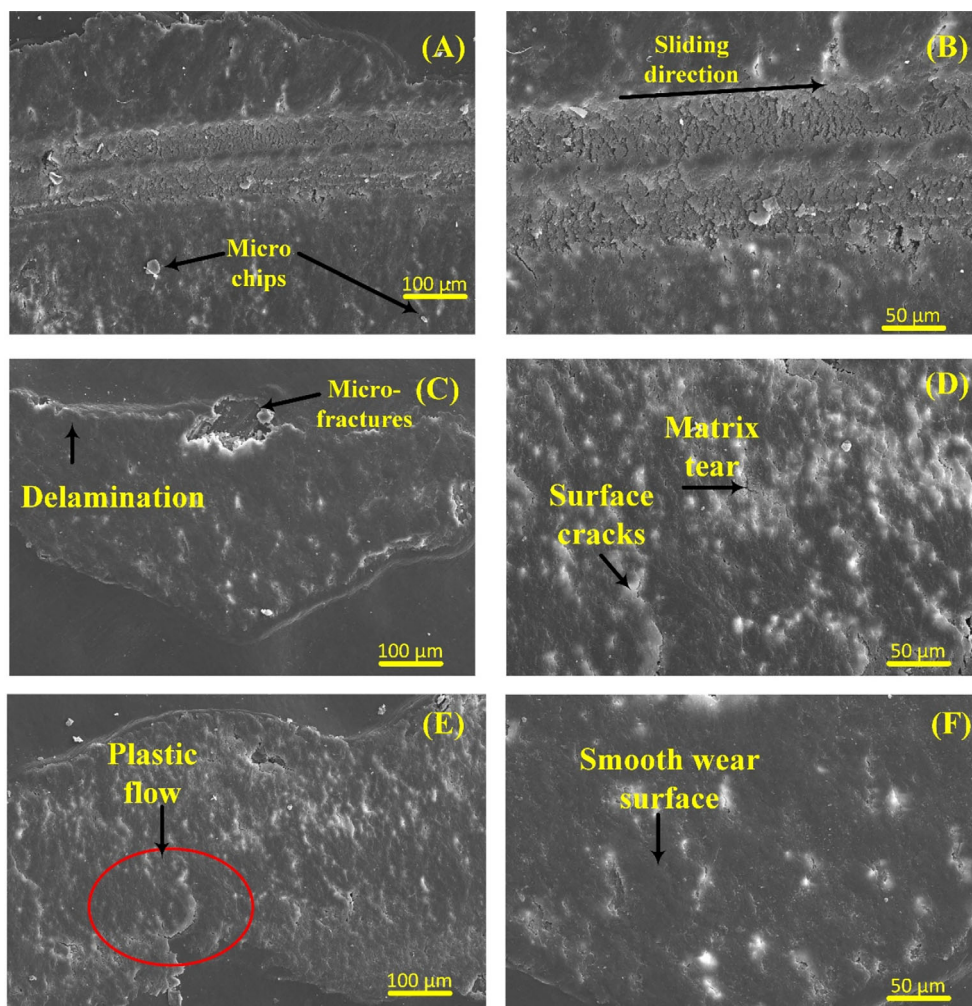
FIGURE 7 SEM images of worn samples depending on filler type; (A) Neat 500 $\times$ magnification, (B) Neat 1000 $\times$ magnification, (C) WC-filled 500 $\times$ magnification, (D) WC-filled 500 $\times$ magnification, (E) ZnO-filled 500 $\times$ magnification, (F) ZnO-filled 1000 $\times$ magnification.

are observed along the wear direction. The lack of hard inorganic particles in epoxy caused the surface to be unable to resist shear forces, leading to increased matrix detachment. Due to the low applied load, the fiber layer was not exposed, but the matrix layer was severely deformed. In Figure 7B, the worn surface of the WC-filled hybrid composite shows matrix softening, but very little detachment is observed. The high hardness of the WC filler and its strong bonding with the matrix minimized matrix damage and contributed to improved wear resistance. Fatigue wear in the matrix resulted in occasional cracks and matrix tears. Mohan et al. observed, through SEM images, that in WC-filled GFRC composites, compared to neat composites, there was less matrix detachment and fiber breakage. They stated that WC fillers strengthen the bond between fibers and the matrix, enhancing abrasive resistance.⁹² In Figure 7C, it is seen that the surface deformations in the ZnO-filled composite are more pronounced compared to the WC-filled composite. Matrix fragmentation and irregular pore formations with varying depths are identified in

specific regions along the wear path, and the surface roughness is more evident. It is observed that the penetration of the counter surface into the matrix layer is more significant. The lower hardness of ZnO compared to WC reduces its load-bearing capacity and increases wear loss. However, compared to the neat composite, the matrix loss is less, and the wear track width is narrower. The contribution of nano-fillers to the formation of tribo-surfaces in the composites ensures that their wear performance is superior to that of neat composites. Therefore, filled composites are more resistant to thermal and mechanical effects. Carrion et al. reported that in polycarbonate (PC)/ZnO composites, the inclusion of ZnO reduced the wear track width and improved tribological performance. However, they also stated that higher ZnO filler content caused abrasive effects, thus increasing wear loss.⁹⁹

Figure 8 illustrates the impact of increasing filler amount on the tribological performance of WC-filled composites. In Figure 8A, it can be observed that the surface integrity of the composite with 0.5% WC filler is

FIGURE 8 SEM images of worn samples depending on filler ratio; (A) 0.5% WC 500 $\times$, (B) 0.5% WC 1000 $\times$, (C) 1% WC-filled 500 $\times$, (D) 1% WC-filled 1000 $\times$, (E) 1.5% WC-filled 500 $\times$, (F) 1.5% WC-filled 1000 $\times$.



more considerably degraded in contrast to the composites with 1% and 1.5% WC fillers. It is noted that microchips are more abundant, and the formations of pits and surface cracks are deeper. In Figure 8B,C, it has been determined that as the WC filler ratio increases, the appearance of the worn surface becomes smoother. With the increasing WC filler content, it can be concluded that the tribo-surface formation across the entire surface results in less damage from the counter pin to the matrix layer. The increase in filler content leads to stronger physical/chemical interactions between fiber/matrix and filler, which improves load transfer capabilities, increases wear resistance, and reduces matrix wear. In Figure 8C, the absence of deep grooves and the soft appearance of the surface, where plastic flow occurs instead of matrix fracture, can be attributed to the inclusion of hard particles within the matrix, which makes the matrix denser and harder, thereby preventing more severe wear. The surface profiles of the 1% and 1.5% WC-filled composites are found to be similar, with matrix debris being rare.

3.7 | Impact strength

Impact resistance can be expressed as a material's ability to resist fracture under high-speed stress. The impact behavior of composites is closely related to their toughness, which is influenced by interfacial and interlaminar stresses. Figure 9 compares the impact resistance results of unfilled and filled hybrid composites with different filler contents of ZnO and WC. It is observed that the ZnO and WC fillers significantly increase the impact resistance of jute/flax composite. The impact resistances of neat, 0.5% ZnO-filled, and 0.5% WC-filled composites are found to be 4.5, 7.9, and 9.1 J/m², respectively. It is understood that an increase in ZnO and WC filler content also enhances impact resistance. Increasing the WC filler content from 0.5% to 1% increased the impact resistance by 6%. The lowest impact resistance was observed in the neat composite, while the highest impact resistance was found in the 1.5% WC-filled composite. It is determined that the WC filler is more effective than the ZnO filler in dissipating impact energy. When epoxy

matrix composites are subjected to impact conditions, the impact energy first deforms the matrix, then separates the matrix-reinforcement interface, and finally transfers to the filler material. The nanometer size of WC and ZnO filler particles ensures a larger surface area, which absorbs impact energy transferred to epoxy, leading to improved impact resistance. The compatibility of the matrix and fillers, along with strong bonding, further enhances the energy absorption capacity.¹⁰⁰ Additionally, the ability of the ZnO filler to hinder crack propagation during impact is another factor that enhances impact resistance.^{101,102} In another study investigating the mechanical performance of WC-filled composites, it was noted that impact strength was improved with the incorporation of WC filler. The study emphasized that WC fillers significantly reduce crack formation and propagation by filling voids within the

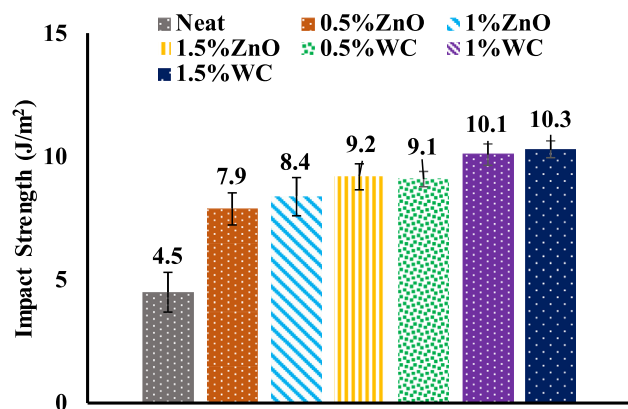


FIGURE 9 Effect of filler content on the impact strength of composites.

matrix and enhance impact resistance by strengthening the bonds between the matrix and fibers.⁶³ Şomoghi et al. reported that ZnO filler particles improve the impact resistance of polyester epoxy, but when the content exceeds 5%, a declining trend in impact resistance is observed.¹⁰³

Figure 10 illustrates SEM analysis of neat jute-flax hybrid composites and WC, ZnO-filled jute-flax hybrid composites fractured under impact load. Deformations on the epoxy matrix, voids, crack initiations at the void ends, cracks, fiber separations, and fractures without any tendency for elongation in the fibers were observed in all neat and WC, ZnO-filled jute-flax composites. As indicated in Figure 10B the high impact resistance obtained in WC and ZnO-filled hybrid composites compared to neat hybrid composites is attributed to plastic deformations and improved void filling by WC and ZnO-filled nanoparticles.¹⁰⁴ Moreover, as illustrated in Figure 10A, weak adhesion bonding between jute-flax fibers and matrix resulted in lower impact resistance of neat jute-flax hybrid composites compared to those filled with WC and ZnO nanoparticles.¹⁰⁵ Jute and flax fibers were separated from the composite matrix due to the impact load and poor adhesion between jute-flax fibers and matrix for neat jute-flax hybrid composites.¹⁰⁶ A close relation was recognized between the matrix deformation and the interface bonds in the jute-flax hybrid composites. The weakness of the interfacial bonds facilitated the propagation of matrix cracks, ultimately leading to composite fracture under impact loading.¹⁰⁷ WC and ZnO nanoparticles filled the voids between fibers and matrix and provided higher impact resistance by enhancing energy absorption of hybrid composites. In WC and

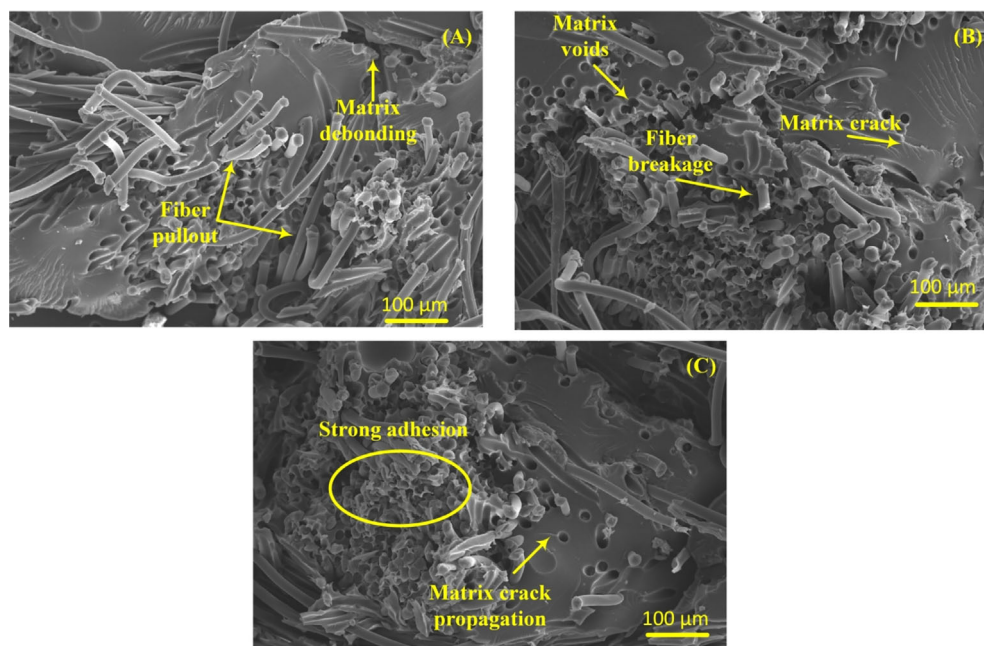


FIGURE 10 SEM photos depicting the fracture surface of impact specimens (A) Neat, (B) ZnO-filled, (C) WC filled.

ZnO-filled hybrid composites, matrix crack initiations were observed less frequently. The impact test results presented in Figure 9 confirm the SEM observations. The high deformations in neat hybrid composites compared to WC and ZnO-filled hybrid composites indicate a lower amount of energy absorbed.¹⁰⁸

3.8 | Prediction of wear performance using neural computation

Figure 11 presents regression obtained from ANN analysis for the COF value derived from experimental results, along with the experimental/predicted graph, K-fold

training and test performance, error histogram, and performance graph. In Figure 11A, when examining the regression curve for COF, it was found that the R-squared values for training, test, validation, and overall results were 0.9848, 0.9979, 0.9116, and 0.9809, respectively. The clustering of predicted values around the ideal target line indicates a high correlation and suggests that the network has established a strong interlink. Figure 11B compares the COF values obtained experimentally and those predicted by a model. An additional orange line represents the error values, indicating the difference between experimental and predicted COF values. Both curves follow a very similar trend, indicating that the predictive model captures the overall behavior of the experimental

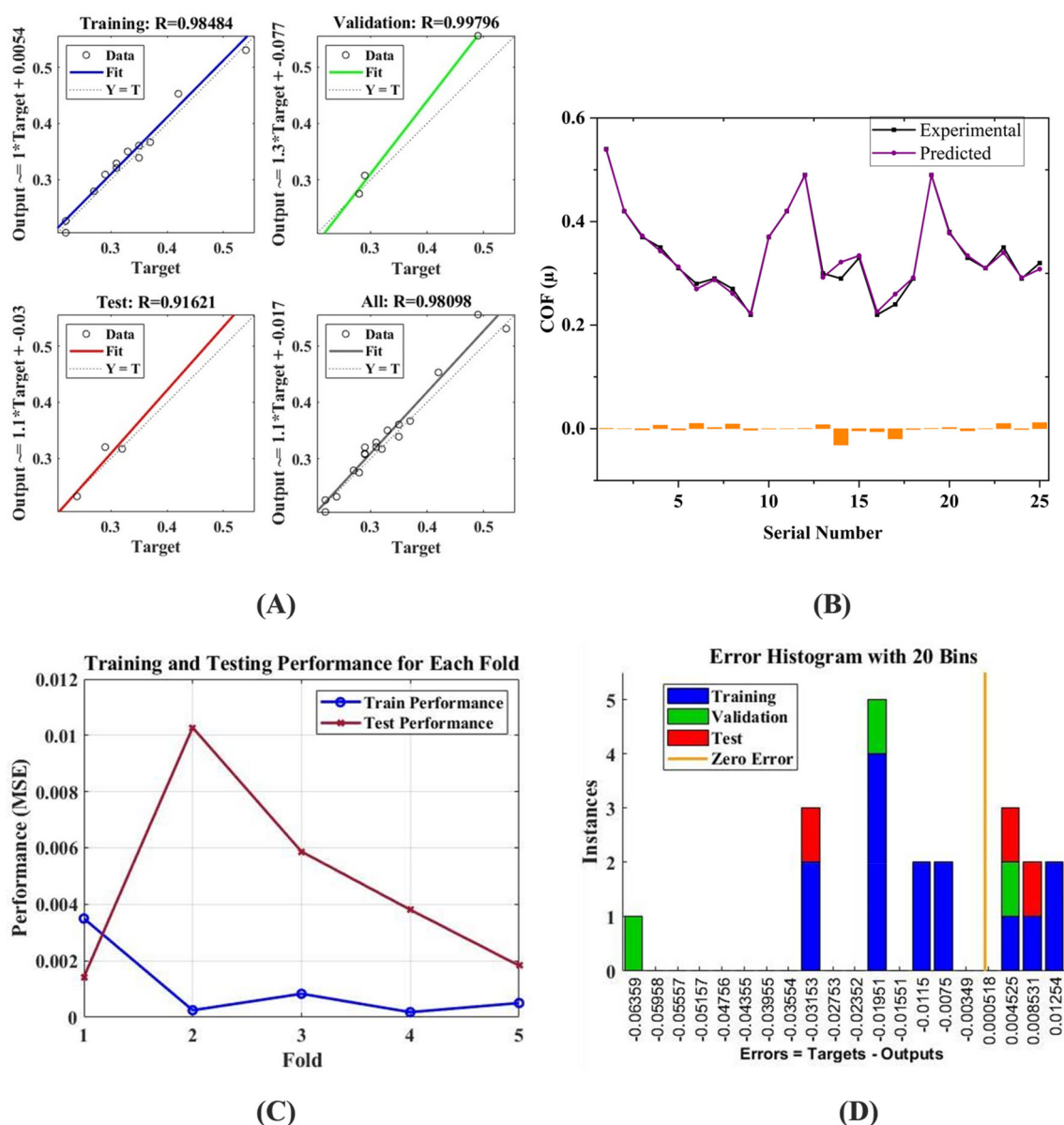
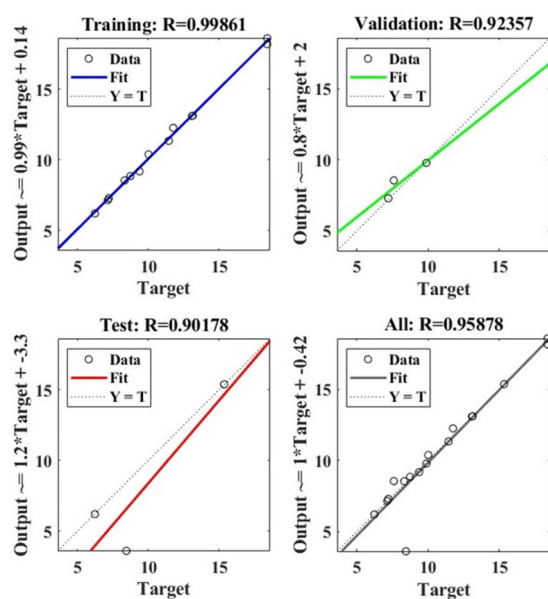


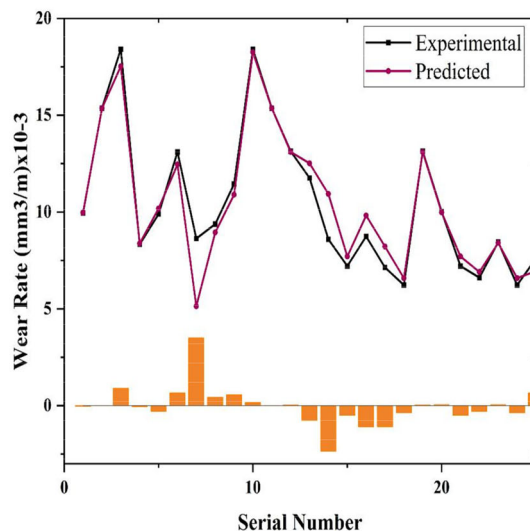
FIGURE 11 ANN performance of COF; (A) Regression plot, (B) Experimental/predicted results, (C) Training/test performance for each fold, (D) Error histogram.

data with good accuracy. The error values are very close to zero for most data points, demonstrating that the model's predictions are highly reliable. The maximum deviation appears to be minor and is within an acceptable range. No drastic discrepancies suggest that the data is of high quality and the model parameters are tuned effectively. The strong agreement between predicted and experimental COF values implies that the model could be confidently applied in practical scenarios to estimate COF without extensive experimental measurements. Figure 11C represents the training and testing performance for each fold in a K-fold cross-validation experiment. The training performance is consistently low

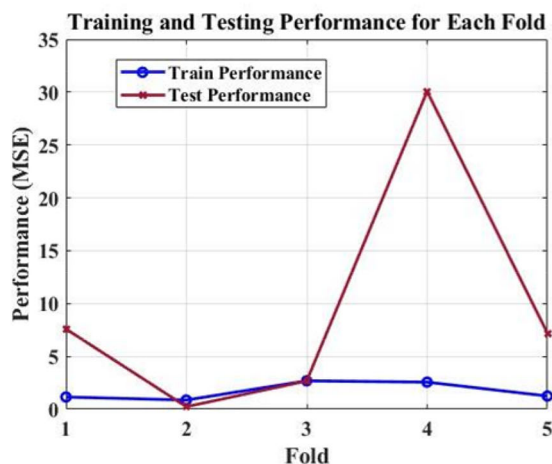
across all folds, with values close to zero. This indicates that the model fits the training data very well for all folds, demonstrating effective learning during the training process. Fold 2 has a significantly higher mean squared error (MSE), indicating poorer generalization on this fold compared to others. Other folds exhibit lower MSE values, with the lowest testing error observed in Fold 5. Figure 11D illustrates an error histogram of COF values. A concentration of errors near zero implies better model performance. The concentration of errors around 0.01951 for training and validation datasets indicates a systematic bias or pattern. These results suggest that the model has acceptable performance but may require further optimization to



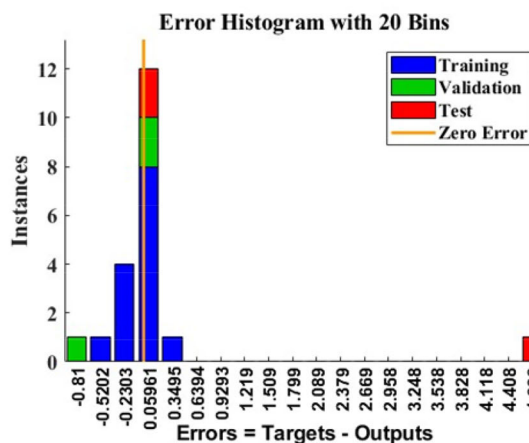
(A)



(B)



(C)



(D)

FIGURE 12 ANN performance of wear rate; (A) Regression plot, (B) Experimental/predicted results, (C) Training/test performance for each fold, (D) Error histogram.

reduce error magnitude and improve generalization. The best validation performance is achieved at epoch 2, with an MSE of 0.0015458 suggests this is the optimal stopping point for training to avoid overfitting. Kumar and Rajmohan have developed an ANN method to predict the wear behavior of FRC. In the ANN approach utilizing LM algorithm, the predicted COF and mass loss values closely matched the experimental results. They reported that the overall correlation coefficient for COF was 0.9790.¹⁰⁹ Figure 12A shows scatter plots comparing the model outputs (predictions) against the targets. The high R-value for the training dataset indicates excellent performance on the training data. The model exhibits strong predictive capability for the training dataset but demonstrates reduced accuracy and slight bias in the validation and test datasets. The combined dataset's R-value (0.95878) confirms that the model performs well overall and is reliable for most scenarios. Figure 12B compares the experimental and predicted wear rates of samples. Residuals are relatively small, reflecting minimal deviation between experimental and predicted values. Figure 12C illustrates the MSE for both training and testing performance over 5 folds. The MSE values for training remain relatively low and stable across all folds, fluctuating slightly between approximately 1 and 10. This indicates that the model achieves a good fit on the training data. Folds 1, 2, 3, and 5 exhibit much lower testing errors (below 10 MSE), indicating better generalization. Figure 12D shows the error histogram of wear rates. Most instances fall within the central bins around 0.05961 with an error close to zero. The training data exhibits a high frequency of errors near zero, suggesting that the model fits the training data effectively. Best validation performances of wear rate over five epochs highlight the point of optimal validation performance (0.30935) achieved at epoch 2. Shai-sundaram et al. analyzed the thermal and wear performance of alumina-filled FRC using an ANN approach. They further validated their findings through cross-validation analysis. Their observations indicated that the ANN model developed for predicting the COF and specific wear rate demonstrated high accuracy and reliability. Additionally, they reported a decreasing trend in the MSE for both training and cross-validation scores as the number of training samples increased.¹¹⁰ Similar results have also been reported in ANN studies related to the wear performance of FRC with an epoxy matrix.^{111–113}

The error rates for COF prediction range between 0% and 8%. However, most of the predicted COF error rates remain below 5%. The highest deviation, 8.28%, was observed in the 17th experimental data point. For wear rate predictions, the minimum and maximum deviation rates were determined as 0.57% and 8.51%, respectively. The low error rates observed for both

COF and wear rate confirm that the network model has been successfully trained.

4 | CONCLUSIONS

In this study, mechanical properties and tribological performance of jute/flax composites with various fillers and contents, produced by the hand lay-up, were experimentally investigated, and wear performance was analyzed using ANN. Both WC and ZnO-filled hybrid composites enhanced the tensile strengths. The highest tensile strength in the WC-filled composite was 3570 N at a 1% filler content, while in the ZnO-filled composite, it was 3148 N at a 1.5% filler content. The highest flexural strengths in the WC and ZnO-filled composites were observed at a 1% filler content. The 1% WC filler increased the flexural strength of the hybrid composite from 59.6 to 72.8 MPa. The increase in tensile and flexural strengths is attributed to the strong interaction between nanoparticle fillers and the matrix, which delays matrix fracture and crack propagation while enhancing load transfer capacity. The occurrence of dense fiber fractures nearby the matrix and fibers pullouts further from the matrix indicates that both the fibers and epoxy resin were resistant to tensile stress, demonstrating sufficient compatibility. Molecular configuration of neat and filled composites was examined using ATR-FTIR, which confirmed the interaction between the epoxy and the fillers. In wear tests, both WC and ZnO fillers contributed to the reduction of COF and wear loss of neat composite by acting as a barrier between matrix and counter surface. An increase in the ZnO filler ratio from 0.5% to 1.5% reduced COF value from 0.38 to 0.31. An increase in sliding distance raised the COF values of both filled and neat composites. It was found that the increase in load led to higher mass loss, while the addition of fillers reduced it. The lowest material loss, 0.32 mg, was observed in the 1.5% WC-filled composite under a 3 N load. Wear rate of the neat hybrid composite decreased from 13.14 to 8.45 mm³/m with 0.5% WC filler, while with the same amount of ZnO filler, it decreased to 10.02 mm³/m. SEM images of the worn surfaces showed that ZnO and WC fillers improved tribological performance by reducing matrix deformation compared to neat composites, and they were more effective in preserving surface integrity. The formation of a durable interface adhesion between nano-sized ZnO and WC fillers and matrix allows effective absorption of impact energy, enhancing the impact strength of composites. It was noted that WC filler was more effective than ZnO, and performance declined at excessive filler concentrations. The highest impact resistance was observed in the 1.5% WC-filled

composite, with a value of 10.3 J/m². The prediction results obtained from the ANN model developed for estimating COF and wear rate are consistent with experimental results, confirming the superior performance of the well-trained network.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

ORCID

Mehmet Emin Demir  <https://orcid.org/0000-0001-9630-6378>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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